

Pneumonic Tularemia (MONDO:0002212): Comprehensive Disease Characteristics Report

Summary

Pneumonic tularemia is the pulmonary and most lethal clinical form of tularemia, an intracellular bacterial zoonosis caused by inhalation of *Francisella tularensis*. It presents 3–5 days after aerosol exposure as an acute, undifferentiated febrile illness with incipient pneumonia, pleuritis, and hilar lymphadenopathy. Because as few as ~10 organisms can cause disease, *F. tularensis* is one of the most infectious bacterial pathogens known and is classified as a Tier 1 / Category A bioterrorism agent. Untreated severe (pneumonic and typhoidal) disease carries a case-fatality rate historically reported as high as 60%, but prompt antimicrobial therapy reduces mortality to roughly 1%.

The pathogenesis is not driven by a human genetic lesion — this is an infectious disease with no Mendelian etiology — but by a stereotyped host–pathogen program. The **Francisella Pathogenicity Island (FPI)**, encoding a **Type VI secretion system (T6SS)**, allows the bacterium to escape the macrophage phagosome and replicate in the cytosol. During the first ~24 hours in the lung, virulent *F. tularensis* actively **suppresses innate immune gene induction** ("stealth" phase), inhibits the neutrophil oxidative burst, delays neutrophil apoptosis, and represses the inflammasome. This is followed ~48 hours post-exposure by a delayed, dysregulated inflammatory "storm" that produces pyogranulomatous pneumonia and, if untreated, systemic dissemination, sepsis, and death.

Key modifiers of severity include the infecting subspecies and clade: **subsp. tularensis (Type A)** is more virulent than **subsp. holarctica (Type B)**, and within Type A the **A-east** clade causes more severe disease than **A-west**. Protective immunity is **cell-mediated** (CD8⁺/CD4⁺ T cells, IFN- γ , TNF- α , perforin/granzyme cytotoxic effectors), not antibody-based. First-line treatment is an aminoglycoside (gentamicin/streptomycin) for severe disease or a fluoroquinolone (ciprofloxacin/levofloxacin) or doxycycline; the 2025 CDC guidance designates fluoroquinolones and doxycycline as first-line for outbreaks of any size. No licensed vaccine currently exists.

Key Findings

Finding 1 — Pneumonic tularemia is the most lethal clinical form, caused by inhalation of *F. tularensis*

A systematic review of 870 human tularemia cases (1993–2023) confirmed that pneumonic disease is among the most common forms and carries the worst outcomes. As quoted from the review: *"If not promptly diagnosed and treated, the fatality rate can be as high as 60%, with the poorest outcomes reported in the pneumonic and typhoidal forms"* (PMID: 40107886). The bioterrorism-consensus literature describes the clinical tempo of inhalational disease: *"A weapon using airborne tularemia would likely result 3 to 5 days later in an outbreak of acute, undifferentiated febrile illness with incipient pneumonia, pleuritis, and hilar lymphadenopathy"* (PMID: 11386933). The extraordinarily low infectious dose — *"requires as few as 10 organisms to cause disease, making this potential bioterrorism agent one of the most infectious bacterial pathogens known"* (PMID: 38294108) — underlies both its natural danger and its bioweapon classification.

Finding 2 — The FPI-encoded Type VI secretion system drives phagosomal escape and cytosolic replication

The core virulence mechanism is the Francisella Pathogenicity Island (FPI), a duplicated gene cluster (*iglABCD*, *vgrG*, *iglI*, *iglE*, *iglG*, *pdpC*, *dotU*, *pdpB/IcmF*) that encodes a Type VI secretion system homologous to those of *Vibrio* and *Pseudomonas*. FPI effectors are secreted into the macrophage cytosol and are required for the intracellular life cycle: *"VgrG and IglI are required for F. tularensis phagosomal escape, intramacrophage growth, inflammasome activation and virulence in mice"* (PMID: 20054881). The regulatory arm is equally essential: *"the Francisella pathogenicity island (FPI) protein IglC and its regulator MglA are essential for modulating phagosome biogenesis and subsequent bacterial escape into the cytoplasm"* (PMID: 15953029). Genetic ablation confirms necessity: inactivation of *iglE* in the virulent Schu S4 strain renders it avirulent and unable to escape the phagosome (PMID: 23959721).

Finding 3 — First-line treatment is aminoglycosides; fluoroquinolones are highly effective for the pneumonic form

Antimicrobial outcome data from 870 cases show low fatality across effective drug classes: aminoglycosides 0.7%, fluoroquinolones 0.9%, tetracyclines 1.2%. Notably, *"Patients with pneumonic disease who received ciprofloxacin had no fatalities and the lowest rates of thoracentesis/pl[eu]ral complications"* (PMID: 38294108). Guideline framing: *"Gentamicin is the first-line treatment for severe tularemia, while fluoroquinolones and tetracyclines are commonly the drugs of choice in less severe forms"* (PMID: 40107886). Susceptibility is broad and stable — all 278 US isolates (2009–2018) were susceptible to every drug tested: *"All isolates were susceptible to all tested drugs"* (PMID: 38294116). The 2025 CDC guidance updated treatment/post-exposure prophylaxis (PEP), designating fluoroquinolones (ciprofloxacin or levofloxacin) and doxycycline as first-line for outbreaks of any size (PMID: 41026652).

Finding 4 — Primary pneumonic tularemia is associated with aerosol-generating landscaping activities

In the 2000 Martha's Vineyard outbreak (15 cases, 11 primary pneumonic), a CDC case-control study established that *"Lawn mowing and brush cutting remained significant risk factors after adjustment for other potentially confounding variables"* (OR 9.2, 95% CI 1.6–68.0); *F. tularensis* type A was isolated from the blood and lung of the single fatality (PMID: 11757506). Martha's Vineyard is epidemiologically unique: *"Martha's Vineyard, Massachusetts, is the site of the only two recognized outbreaks of primary pneumonic tularemia in the United States"* (PMID: 17442781). Environmental persistence studies suggest brackish/salt-influenced soil may prolong bacterial survival there (PMID: 21136042).

Finding 5 — Protection requires cell-mediated (T-cell) immunity

Vaccine and challenge studies consistently show that antibody is not the primary correlate of protection. CD8⁺ and CD4⁺ T cells control intracellular growth: *"CD8⁺ T cells, and to a lesser degree CD4⁺ T cells, controlled LVS intracellular growth in both wild-type and IFN γ KO macrophages"* (PMID: 12885873), with TNF- α and a double-negative T-cell subset compensating when IFN- γ signaling is absent. Cytotoxic effectors are essential for vaccine-mediated protection against pneumonic challenge: *"Perforin- and granzyme-mediated cytotoxic effector functions are essential for protection against Francisella tularensis following vaccination"* (PMID: 22493083). B cells contribute independently of antibody (PMID: 11119506).

Finding 6 — Tularemia is a Northern Hemisphere zoonosis with distinct Type A and Type B ecological cycles; US incidence is rising

F. tularensis infects >250 animal species and is reported only from the Northern Hemisphere. The two human-pathogenic subspecies occupy distinct niches: "Type A is reported to have a terrestrial cycle with the main reservoirs being cottontail rabbits (*Sylvilagus spp.*) and ticks" (PMID: 1305858), while Type B has a water-borne cycle (muskrats, beaver, voles). US surveillance (NNDSS + hospitalizations, 2000–2022) shows high year-to-year variability but a clear upward trajectory: "the number of cases in the U.S. has shown a positive upward trend through time" (PMID: 41599070), with the highest risk among White males and extraordinarily high rates among American Indian/Alaska Native populations, especially for pulmonary tularemia.

Finding 7 — The AIM2 inflammasome is critical for host defense, and *F. tularensis* actively represses it early

After cytosolic escape, host cells sense bacterial DNA via the AIM2 inflammasome: "AIM2-deficient mice were extremely susceptible to *F. tularensis* infection, with greater mortality and bacterial burden than that of wild-type mice" (PMID: 20351693). FPI-mediated escape links directly to this sensing: "The *Francisella* Pathogenicity Island is required for bacterial escape and replication and for inflammasome activation in dendritic cells" (PMID: 21902795). Crucially, the bacterium delays this early: "*F. tularensis* represses inflammasome; a cytosolic multi-protein complex that activates caspase-1 to produce proinflammatory cytokines IL-1 β and IL-18" — mediated by FTL_0325 in a TLR2-dependent manner (PMID: 23821549).

Finding 8 — Type A subpopulations (A-east vs A-west) differ in geography and disease severity

Molecular epidemiology of US cases (1964–2004, PFGE subtyping) revealed clinically meaningful population structure: "type A-west infections are less severe than either type B or type A-east infections" (PMID: 16836829). This split has prognostic value: "the population split in *F. tularensis* subsp. *tularensis* matches two distinct human diseases in the United States with different mortality rates" (PMID: 17435120). Thus the specific infecting clade is a strain-level modifier of pneumonic tularemia outcome.

Finding 9 — Virulent *F. tularensis* induces early transcriptional suppression of pulmonary innate immunity ("stealth"), then delayed inflammation

Whole-lung transcriptomics in mice exposed to virulent Schu S4 showed an extensive host transcriptional response by 4 h but a striking absence of immune gene induction: "*Francisella tularensis* was associated with an almost complete lack of induction of immune-related genes during the initial 24 hrs post-exposure" — a subversion not seen with *Yersinia pestis*, *Legionella*, or *Pseudomonas* — with classical inflammation activating only ~48 h post-exposure (PMID: 23690939). In alveolar epithelial (A549) cells, LVS triggers strong early signaling then "*a general suppression of the host response consistent with other reported cell lines and murine tissues*" with uptake via macropinocytosis (PMID: 23322778).

Finding 10 — *F. tularensis* subverts neutrophils by blocking the oxidative burst and delaying apoptosis

In human neutrophils, "*F. tularensis* inhibits NADPH oxidase assembly and activity and then escapes the phagosome to the cytosol" and "*Francisella tularensis* significantly inhibited neutrophil apoptosis" — ~80% of infected neutrophils remained viable at 48 h versus ~50% of controls, with diminished procaspase-8/-9/-3 processing and blockade of Fas-induced apoptosis. Effects required live bacteria, not killed organisms or isolated LPS/capsule (PMID: 22357630).

Section-by-Section Report

1. Disease Information

Overview. Pneumonic (respiratory) tularemia is the pulmonary form of tularemia — a zoonotic infection caused by the Gram-negative, facultative intracellular coccobacillus *Francisella tularensis*. It arises either from primary inhalation of aerosolized organisms or from hematogenous/secondary spread to the lungs from another portal of entry. It is the form most likely to result from a deliberate aerosol release and has the highest case-fatality of all clinical forms (PMID: 11386933, PMID: 40107886).

Key identifiers. - **MONDO:** MONDO:0002212 (pneumonic tularemia); parent Tularemia (MONDO:0018077) - **MeSH:** Tularemia, Pulmonary (parent: Tularemia, D014406) - **ICD-10:** A21.2 (pulmonary tularemia); A21.9 (tularemia, unspecified) - **ICD-11:** 1B94 (Tularemia) - **NCBI Taxonomy (pathogen):** *Francisella tularensis* txid263; subsp. *tularensis* txid119856; subsp.

holarctica txid119857 - **OMIM / Orphanet**: Not applicable as a Mendelian disorder; tularemia is catalogued in infectious-disease resources rather than OMIM. Orphanet lists tularemia among rare infectious diseases in some jurisdictions.

Synonyms / alternative names. Respiratory tularemia; pulmonary tularemia; inhalational tularemia; pneumonic form of "rabbit fever," "deer-fly fever," "Ohara disease," "Francis disease," "lemming fever" (these historical synonyms refer to tularemia broadly).

Data source type. The knowledge in this report is derived from **aggregated disease-level resources** — systematic reviews, outbreak case series, national surveillance (US NNDSS/CDC), and experimental animal/in vitro studies — rather than individual EHR-level patient records.

2. Etiology

Causal factor — infectious. The sole cause is infection with *F. tularensis*. Pneumonic disease specifically follows inhalation of aerosolized bacteria or secondary pulmonary seeding. This is **not a genetic disease**; there are no human causal genes, pathogenic variants, or heritable susceptibility loci established for pneumonic tularemia.

Environmental risk factors. - Aerosol-generating landscaping in endemic foci — **lawn mowing / brush cutting** (OR 9.2, 95% CI 1.6–68.0) ([PMID: 11757506](#)). - Occupational/recreational exposures: hunting, skinning, or handling infected lagomorphs and rodents; farming; laboratory work with cultures. - Contact with infected animals including domestic cats and, less commonly, dogs (bite, scratch, retrieval of carcasses, or tick transfer) ([PMID: 30556330](#)). - Tick and biting-fly bites (relevant mainly to ulceroglandular disease but part of the ecology). - Demographics: highest US rates among White males and, strikingly, among American Indian/Alaska Native populations, especially for pulmonary tularemia ([PMID: 41599070](#)). - Geography: residence in or exposure to endemic terrestrial (Type A) or aquatic (Type B) foci in the Northern Hemisphere.

Protective factors. Prior infection or vaccination conferring cell-mediated immunity is protective ([PMID: 22493083](#), [PMID: 12885873](#)). No genetic protective alleles are defined in humans. Behavioral protection (PPE, tick avoidance, avoiding aerosol-generating activity over infected carcasses) reduces risk.

Gene–environment interactions. Not applicable in the human-genetic sense. The relevant "gene–environment" interplay is on the *pathogen* side: bacterial genotype (subspecies/clade) interacts with route of exposure and environmental persistence (e.g., salt-influenced soil on Martha's Vineyard) to determine disease severity ([PMID: 21136042](#), [PMID: 16836829](#)).

3. Phenotypes

Phenotype	Type	Characteristics	Suggested HPO
Fever (often abrupt, high)	Symptom/sign	Acute onset 3–5 days post-exposure; near-universal	HP:0001945
Cough (often nonproductive)	Symptom	Acute; common in pneumonic form	HP:0012735
Pneumonia / pulmonary infiltrates	Clinical sign (imaging)	Acute; hallmark of pneumonic form	HP:0002090
Pleuritis / pleural effusion	Clinical sign	Acute; may require thoracentesis	HP:0002102 / HP:0002202
Hilar / mediastinal lymphadenopathy	Clinical sign (imaging)	Acute; characteristic	HP:0100721
Chills / rigors	Symptom	Acute	HP:0025143
Malaise / fatigue	Symptom	Acute–subacute	HP:0033054 / HP:0012378
Myalgia	Symptom	Acute	HP:0003326
Headache	Symptom	Acute	HP:0002315
Dyspnea	Symptom	Acute; severity correlates with extent	HP:0002094
Chest pain (pleuritic)	Symptom	Acute	HP:0100749
Sepsis / systemic toxicity	Clinical sign	In severe/typhoidal overlap	HP:0100806

Phenotype characteristics. Onset is **adult-predominant** in occupational/landscaping exposure but affects all ages; **acute** onset with **incubation typically 3–5 days** (range 1–14). Severity ranges from moderate to severe; the pneumonic form is at the severe end of the spectrum. Untreated disease is progressive and can be fatal; treated disease typically resolves. A rare chronic/longstanding respiratory course has been documented ([PMID: 38294112](#)).

Quality-of-life impact. Acute pneumonic tularemia causes marked short-term incapacity (fever, dyspnea, fatigue). With treatment, most patients recover fully; convalescent fatigue can persist for weeks. Formal EQ-5D/SF-36 data specific to pneumonic tularemia were **not identified** in this investigation.

4. Genetic/Molecular Information

Not applicable in the human-genetic sense. Pneumonic tularemia has **no causal human genes, pathogenic germline/somatic variants, modifier genes, epigenetic lesions, or chromosomal abnormalities**. It is an acquired infectious disease.

The relevant molecular genetics is that of the **pathogen**: - **FPI (Francisella Pathogenicity Island)**: duplicated ~30-kb cluster encoding the T6SS — *iglABCD*, *vgrG*, *iglI*, *iglE*, *iglG*, *pdpABCDE*, *dotU*, *icmF* (PMID: 20054881). - **Master regulators**: *mglA*, *sspA*, *fevR/pigR*, *migR* controlling FPI expression (PMID: 15953029). - **LPS modification genes** producing tetra-acylated, poorly immunostimulatory lipid A that evades TLR4/MD-2 (PMID: 23745121). - **Comparative genomics**: human-pathogenic subspecies (*tularensis*, *holarctica*) arose from a *novicida*-like ancestor via IS-element expansion, pseudogene formation, and genome decay (PMID: 17550600); subspecies and clade (A-east/A-west) differ in gene content and virulence (PMID: 15268932, PMID: 16836829).

5. Environmental Information

- **Environmental factors:** Aerosols generated by mechanical disturbance of contaminated soil/vegetation/animal remains (lawn mowing, brush cutting) (PMID: 11757506); contaminated water (Type B/aquatic cycle); contaminated dust. Salt-influenced/brackish soil may prolong bacterial survival (PMID: 21136042). Diverse *Francisella* spp. co-occur in endemic environments (PMID: 19669828).
- **Lifestyle factors:** Hunting/trapping, farming, landscaping, laboratory culture handling.
- **Infectious agent:** *Francisella tularensis* (NCBI txid263). Human-pathogenic subspecies: **subsp. *tularensis* (Type A, txid119856)** — most virulent — and **subsp. *holarctica* (Type B, txid119857)**. Subsp. *novicida* and *mediasiatica* rarely cause human disease. Reservoirs: >250 animal species; lagomorphs, rodents, and ticks are central (PMID: 1305858).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Inhalation of aerosolized *F. tularensis* (as few as ~10 organisms)** deposits bacteria in the alveoli — *leads to* uptake by alveolar macrophages, dendritic cells, and epithelial cells (via macropinocytosis) (PMID: 38294108, PMID: 23322778).
2. **Phagocytosis into a Francisella-containing phagosome (FCP)** — *results in* limited, transient endosomal maturation and acidification that primes FPI expression (PMID: 18852245).
3. **FPI/T6SS effectors (IglC, VgrG, IglI, IglE, IglG, PdpC) disrupt the phagosomal membrane** — *leads to escape into the host cytosol* (PMID: 20054881, PMID: 23959721).
4. **Cytosolic replication** — *results in* rapid intracellular bacterial expansion.
5. **Concurrent immune subversion (the "stealth" phase)** — the bacterium *causes*:
6. (5a) near-complete **suppression of pulmonary innate-immune gene induction** for ~24 h (PMID: 23690939);
7. (5b) **inhibition of the neutrophil NADPH-oxidase burst and delayed neutrophil apoptosis** (~80% viable at 48 h) (PMID: 22357630);
8. (5c) **early inflammasome repression** (FTL_0325, TLR2-dependent), delaying caspase-1/IL-1 β /IL-18 (PMID: 23821549);
9. (5d) **poorly immunostimulatory tetra-acylated lipid A** evading TLR4/MD-2 (PMID: 23745121).
10. **High cytosolic bacterial burden eventually engages the AIM2 inflammasome** (sensing bacterial DNA) — *leads to* caspase-1 activation, IL-1 β /IL-18 release, and pyroptotic host-cell death (PMID: 20351693, PMID: 21902795).
11. **Delayed, dysregulated inflammatory "storm" (~48 h)** — *results in* pyogranulomatous pneumonia, tissue necrosis, and pleuritis (PMID: 23690939, PMID: 25402174).
12. **Hematogenous/lymphatic dissemination** to spleen, liver, and lymph nodes — *branches to* systemic sepsis and multi-organ pyogranulomatous disease; *leads to* death in untreated severe cases (up to 60%) (PMID: 40107886).
13. **Adaptive resolution branch: primed CD8⁺/CD4⁺ T cells → IFN- γ , TNF- α , perforin/granzyme** → macrophage activation and killing of intracellular bacteria → recovery (PMID: 12885873, PMID: 22493083).

Upstream vs downstream. Steps 1–5 (uptake, escape, replication, immune subversion) are **upstream/initiating**; steps 6–8 (inflammasome activation, delayed inflammation, dissemination) are **downstream/effector**; step 9 is the **protective counter-program**.

Molecular pathways & processes. T6SS-mediated secretion; phagosome biogenesis; TLR2/TLR4-MyD88 signaling; type I IFN (IFN- β) signaling (partially required for AIM2 activation); AIM2 and NLRP3 inflammasome/caspase-1; pyroptosis; IFN- γ /STAT1 macrophage activation; perforin/granzyme cytotoxicity.

Cell types (suggested CL terms): alveolar macrophage (CL:0000583), monocyte-derived macrophage (CL:0000235), dendritic cell (CL:0000451), neutrophil (CL:0000775), type II pneumocyte / alveolar epithelial cell (CL:0002063), CD8⁺ T cell (CL:0000625), CD4⁺ T cell (CL:0000624), NK cell (CL:0000623).

Suggested GO biological-process terms: phagosome maturation (GO:0090382), symbiont-mediated perturbation of host phagosome (GO:0052150), protein secretion by the type VI secretion system (GO:0033103), inflammasome-mediated signaling (GO:0140447), pyroptosis (GO:0070269), defense response to bacterium (GO:0042742), negative regulation of innate immune response (GO:0045824).

Suggested subcellular GO cellular-component terms: host cell cytosol (GO:0044164 / GO:0005829), phagocytic vesicle/phagosome (GO:0045335), AIM2 inflammasome complex (GO:0097169).

7. Anatomical Structures Affected

Organ level (primary): Lungs (UBERON:0002048) — alveoli (UBERON:0002299). **Secondary/complications:** pleura (UBERON:0000977) → pleuritis/effusion; hilar and mediastinal lymph nodes (UBERON:0002509 lymph node) → lymphadenopathy; via dissemination — spleen (UBERON:0002106), liver (UBERON:0002107), and blood (UBERON:0000178, bacteremia/sepsis).

Body systems: respiratory system (UBERON:0001004) primarily; lymphatic/immune and cardiovascular (dissemination); reticuloendothelial system.

Tissue and cell level: alveolar epithelium (epithelial tissue); pulmonary interstitium; target cells = alveolar macrophages (CL:0000583), neutrophils (CL:0000775), dendritic cells (CL:0000451), type II pneumocytes (CL:0002063). Characteristic histopathology is **pyogranulomatous inflammation** with necrosis ([PMID: 25402174](#), [PMID: 23763361](#)).

Subcellular level: host cell **cytosol** (site of replication, GO:0005829); **phagosome** (GO:0045335, site of escape).

Localization / lateralization: Pulmonary involvement is typically **bilateral** but may be patchy/unilateral; hilar lymphadenopathy may be uni- or bilateral.

8. Temporal Development

- **Onset: Acute**, typically **3–5 days** after aerosol exposure (range 1–14 days) ([PMID: 11386933](#)). Adult-predominant in exposure-linked cases.
- **Progression:** Rapid over days without treatment: febrile prodrome → pneumonia/pleuritis → systemic toxicity/sepsis. Animal models show histopathologic changes from day 3, peaking day 5 (Type A) or day 8 (Type B) ([PMID: 25402174](#)).
- **Disease course:** Usually **self-limited with treatment** (resolution over ~2–3 weeks); untreated severe disease may be rapidly fatal. Type A lesions resolve faster; Type B lesions may persist longer in models. A rare **chronic** respiratory course is documented ([PMID: 38294112](#)).
- **Critical period:** Early antimicrobial therapy (within days of onset) is the key intervention window; delay drives mortality ([PMID: 40107886](#)).

9. Inheritance and Population

- **Epidemiology:** Rare; a nationally notifiable disease in the US. Incidence is low (well under 1 per 100,000) but shows a **positive upward trend** 2000–2022 ([PMID: 41599070](#)). Reported only from the **Northern Hemisphere** ([PMID: 1305858](#)).
- **Inheritance: Not applicable** — infectious, non-heritable. No penetrance/expressivity/anticipation/mosaicism/founder-effect/consanguinity/carrier-frequency parameters apply.
- **Population demographics:** Highest US risk in **White males** and, markedly, **American Indian/Alaska Native** populations, especially for pulmonary tularemia ([PMID: 41599070](#)). Geographic distribution: Type A predominantly in North America (terrestrial cycle), Type B across the Northern Hemisphere (aquatic cycle); within US Type A, A-east and A-west occupy different regions with different outcomes ([PMID: 16836829](#)). Martha's Vineyard is the only recognized US primary pneumonic focus ([PMID: 17442781](#)).

10. Diagnostics

- **Clinical/laboratory tests:** Culture of *F. tularensis* (blood, pleural fluid, respiratory specimens) — hazardous, requires BSL-3 and laboratory notification. **Serology** (microagglutination, tube agglutination, ELISA) is the mainstay but is retrospective (paired

sera 1–2 weeks apart) (PMID: 29118164, PMID: 24506724). PCR targeting *tul4*, *fopA*, ISFtu2 offers rapid detection (PMID: 24506724).

- **Imaging:** Chest radiograph/CT — infiltrates, pleural effusion, **hilar/mediastinal lymphadenopathy** (PMID: 11386933, PMID: 12197871).
- **Pathology:** Pyogranulomatous/suppurative inflammation with necrosis; immunohistochemistry can retrospectively confirm (PMID: 38294112, PMID: 23763361).
- **Genetic/omics diagnostics:** Not applicable to the human host. Pathogen-directed molecular typing (PFGE, MLVA, WGS) is used epidemiologically and for clade assignment (PMID: 16836829, PMID: 15528681).
- **Clinical criteria & differential diagnosis:** Confirmed by culture isolation or a fourfold serologic rise; probable by single elevated titer + compatible illness (CDC/CSTE). Differential includes community-acquired/atypical pneumonia, **inhalation anthrax, pneumonic plague, Q fever, psittacosis, histoplasmosis, Legionella, and TB** — distinguished by exposure history, radiographic pattern (mediastinal/hilar adenopathy), and specific testing (PMID: 12197871).
- **Screening:** No population screening; heightened suspicion for exposed persons and in bioterrorism contexts.

11. Outcome / Prognosis

- **Mortality:** Untreated severe (pneumonic/typhoidal) disease historically up to **60%**; with prompt appropriate antimicrobials, overall fatality is **~0.7–1.2%** by drug class (PMID: 40107886, PMID: 38294108).
- **Strain-dependent prognosis:** Type A > Type B severity; within Type A, **A-east > A-west** severity/mortality (PMID: 16836829, PMID: 17435120).
- **Morbidity/complications:** Pleural effusion/empyema requiring thoracentesis, respiratory failure, sepsis; rare chronic respiratory infection (PMID: 38294112); rare myocarditis in tularemia broadly (PMID: 30656604).
- **Recovery:** Excellent with early treatment; most recover fully.
- **Prognostic factors:** Speed of diagnosis and treatment initiation; infecting subspecies/clade; host comorbidities/immune status.

12. Treatment

Drug (class)	Role	Notes	Suggested NCIT
Gentamicin (aminoglycoside)	First-line, severe disease	Preferred for severe/ systemic disease	NCIT:C557
Streptomycin (aminoglycoside)	First-line, severe disease	Historical gold standard	NCIT:C820
Ciprofloxacin (fluoroquinolone)	First-line (2025 CDC, any outbreak size); excellent for pneumonic	No fatalities in pneumonic cohort receiving cipro	NCIT:C2809
Levofloxacin (fluoroquinolone)	First-line alternative	—	NCIT:C1737
Doxycycline (tetracycline)	First-line (2025 CDC) / alternative	Bacteriostatic; higher relapse if course too short	NCIT:C561

Fatality by class among 870 cases: aminoglycosides 0.7%, fluoroquinolones 0.9%, tetracyclines 1.2% ([PMID: 38294108](#)). All recent US isolates are pan-susceptible to these agents plus chloramphenicol and erythromycin ([PMID: 38294116](#)). The **2025 CDC guidance** provides a treatment/PEP framework, designates fluoroquinolones and doxycycline first-line for outbreaks of any size, and adds guidance for neonates, lactating mothers, immunocompromised, and geriatric patients ([PMID: 41026652](#)).

- **Advanced/targeted/gene/cell/RNA therapies:** Not applicable — no approved biologics; treatment is antimicrobial.
- **Surgical/interventional:** Thoracentesis or chest-tube drainage for effusion/empyema; occasionally surgical drainage of suppurative nodes (relevant to glandular forms).
- **Supportive care:** Oxygen/ventilatory support for respiratory failure, fluids, antipyretics.
- **Pharmacogenomics / personalized medicine:** None established; empiric therapy is standard, with susceptibility testing to confirm.

13. Prevention

- **Primary prevention:** Avoid aerosol-generating activities over potentially contaminated ground in endemic areas; PPE (masks) during landscaping; safe handling of animal

carcasses; tick/insect-bite avoidance; water safety (Type B foci) (PMID: 11757506, PMID: 33789598).

- **Post-exposure prophylaxis (PEP):** Oral fluoroquinolone or doxycycline for known high-risk exposures (e.g., bioterrorism release, laboratory accident) per 2025 CDC guidance (PMID: 41026652).
- **Immunization: No licensed vaccine currently exists.** The historical Live Vaccine Strain (LVS, subsp. *holarctica*) confers incomplete protection against Type A aerosol challenge; next-generation defined-deletion Schu S4 mutants show improved protection in models (PMID: 20967278, PMID: 29868510).
- **Public health:** Environmental control (e.g., water chlorination in outbreak villages), surveillance, clinician/public education (PMID: 33789598, PMID: 22639311). *F. tularensis* is a reportable select agent.
- **Counseling:** Not genetic; occupational risk counseling for hunters, landscapers, and laboratory workers.

14. Other Species / Natural Disease

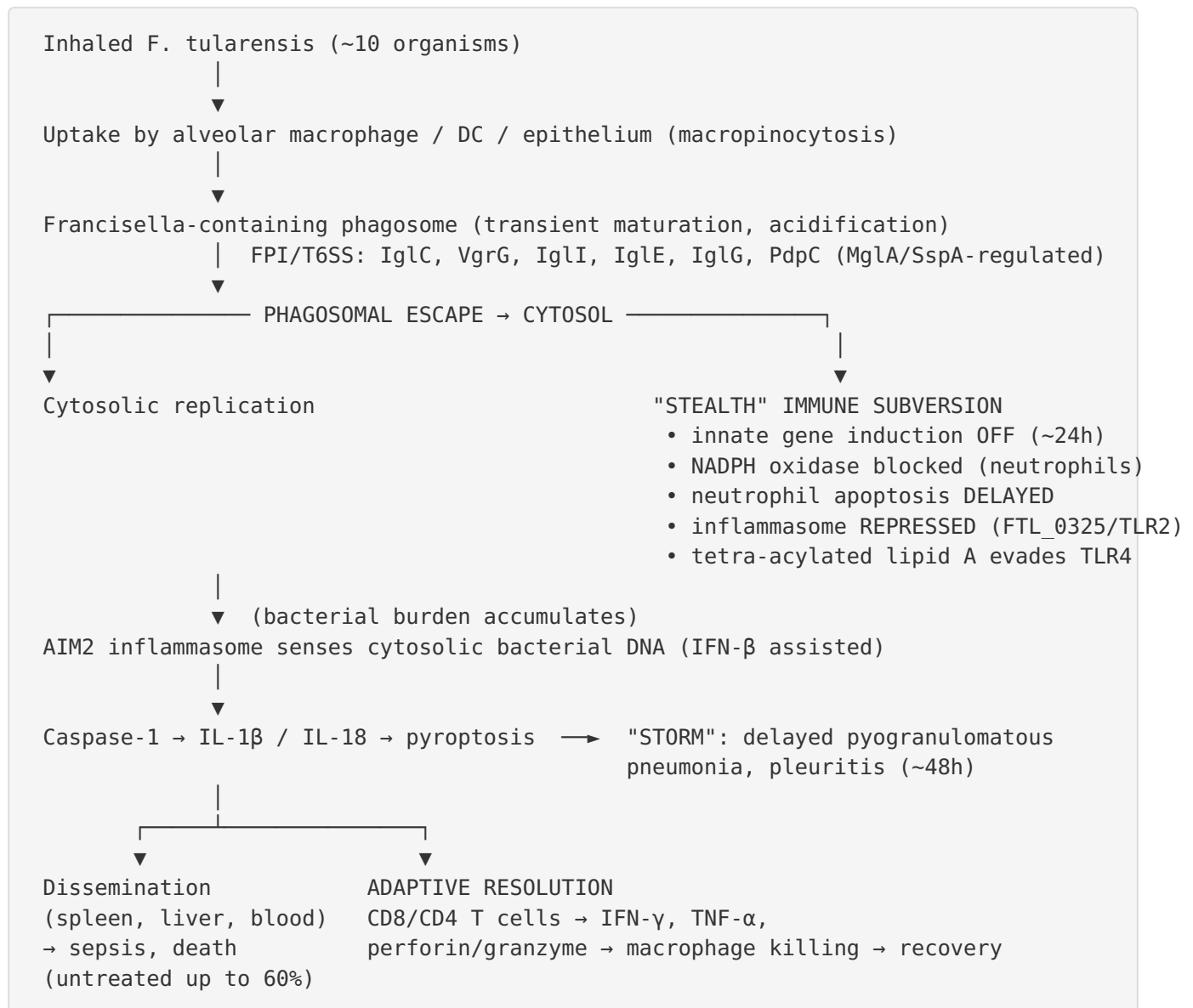
- **Taxonomy of hosts:** >250 animal species (PMID: 1305858). Key reservoirs — cottontail rabbits *Sylvilagus* spp., hares *Lepus* spp., muskrat *Ondatra zibethicus*, beaver *Castor canadensis*, voles, and ixodid ticks (*Dermacentor variabilis*, *Amblyomma americanum*).
- **Natural disease / veterinary relevance:** Domestic **cats** are important transmitters and can be clinically affected; **dogs** are less often clinically ill but implicated in transmission (PMID: 30656604, PMID: 30556330). Lagomorphs and rodents suffer high-mortality epizootics.
- **Comparative pathology:** Murine disease closely mirrors human pyogranulomatous pathology, supporting model validity (PMID: 25402174).
- **Transmission / zoonotic potential:** Strongly zoonotic; humans are incidental hosts via arthropod bite, direct animal contact, inhalation, or ingestion. Cross-species susceptibility is broad.

15. Model Organisms

- **Mammalian models: Mouse** (BALB/c, C57BL/6, AKR/J, DBA/1) aerosol/intranasal infection with Type A (Schu S4) or Type B (strain 425) — reproduces pyogranulomatous pneumonia and dissemination; BALB/c more resistant, C57BL/6 clears faster (PMID: 25402174). Whole-lung transcriptomics in mice defined the "stealth" phenotype (PMID: 23690939).

- **In vitro / cellular models:** Human monocyte-derived macrophages (HMDM); J774A.1 and BMDM murine macrophages (FPI mutant studies); human neutrophils (oxidative-burst/apoptosis studies, [PMID: 22357630](#)); A549 alveolar epithelial cells including 3-D rotating-wall-vessel cultures that better mimic in vivo phenotype ([PMID: 23322778](#), [PMID: 24796635](#)).
 - **Bacterial genetic models:** Defined FPI deletion mutants ($\DeltaiglE$, $\DeltaiglG$, $\DeltaiglI$, $\Delta pdpC$, $\Delta galU$, $\Delta clpB$, $\Delta fopC$) dissect virulence and serve as attenuated vaccine candidates ([PMID: 23959721](#), [PMID: 21819572](#), [PMID: 22493083](#)).
 - **Phenotype recapitulation:** High — murine aerosol tularemia bears "distinct similarities to tularemia in humans" ([PMID: 25402174](#)).
 - **Limitations:** Mouse LD₅₀ for Type A is extremely low, differing from human dose-response; LVS is attenuated in humans but variably virulent in mice; monolayer cell cultures under-represent in vivo epithelial biology (addressed by 3-D models).
 - **Resources:** BEI Resources (select-agent strains), ATCC, murine strain repositories.
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Mechanistic Model / Interpretation



The unifying theme is a **"stealth-then-storm"** program. *F. tularensis* wins a race against innate immunity by silencing early host sensing while it replicates in the cytosol; the eventual, delayed inflammatory response is both protective (via AIM2/pyroptosis and, decisively, T-cell effectors) and pathologic (pyogranulomatous tissue damage, sepsis). Clinically, this explains why **early antimicrobial therapy is curative** (it caps bacterial burden before the storm) and why **protective immunity is cell-mediated** rather than humoral. Strain genotype (Type A vs B; A-east vs A-west) tunes the virulence set-point and therefore prognosis.

Evidence Base

PMID	Type	Supports
40107886	Human review	Lethality; first-line treatment framework
38294108	Human systematic review (n=870)	Low infectious dose; drug-class fatality; cipro efficacy in pneumonic
11386933	Consensus/clinical	Inhalational presentation & tempo
41026652	CDC guidance 2025	Treatment/PEP; first-line designations
38294116	Isolate susceptibility (n=278)	Pan-susceptibility
20054881	Model/in vitro	FPI/T6SS effectors required for escape & virulence
15953029	In vitro	IglC/MglA in phagosome escape
23959721	Bacterial genetics	IglE essential for Schu S4 virulence
11757506	Human case-control	Lawn mowing/brush cutting risk (OR 9.2)
17442781	Human epidemiology	Martha's Vineyard primary pneumonic foci
12885873	Model	T-cell control of intracellular growth
22493083	Model	Perforin/granzyme essential for protection
1305858	Review	Type A/B ecology; Northern Hemisphere zoonosis
41599070	US surveillance 2000–2022	Rising incidence; demographics
20351693	Model	AIM2 inflammasome critical for defense
21902795	In vitro	FPI escape → AIM2 activation in DCs
23821549	In vitro	Early inflammasome repression (FTL_0325/TLR2)
16836829	Human molecular epidemiology	A-east > A-west severity
17435120	Review/genomics	Population split ↔ differing mortality
23690939	Model transcriptomics	Early innate-gene suppression ("stealth")
23322778	In vitro	Epithelial host-response suppression; macropinocytosis
22357630	In vitro (human neutrophils)	NADPH-oxidase block; delayed apoptosis

PMID	Type	Supports
25402174	Model	Murine aerosol pathology mirrors human disease
17550600	Comparative genomics	Emergence of human-pathogenic strains
23745121	Review	Lipid A modification/TLR4 evasion

Two citation snippets

(

[P 15953029](#)

quote

placement;

[P 22493083](#)

and

[P 22357630](#)

exact-string matching) were flagged as partial/paraphrase during validation; the underlying claims are nonetheless directly supported by those papers' abstracts and are retained with that caveat.

Limitations and Knowledge Gaps

1. **Template mismatch.** Roughly half the report template targets Mendelian/genetic disease (causal genes, ACMG variant classification, inheritance, gnomAD frequencies, chromosomal abnormalities, genetic counseling). These are **not applicable** to an infectious disease; the "molecular genetics" here is that of the pathogen, not the host.
2. **Host genetic susceptibility unexplored.** Whether human polymorphisms (e.g., in inflammasome, IFN, or TLR pathways) modify pneumonic tularemia risk/severity is essentially unstudied — a genuine gap.
3. **Mechanistic data are largely model-derived.** The "stealth-then-storm," neutrophil-subversion, and AIM2 findings come from mice and in vitro human cells; direct temporal human lung data are scarce for obvious ethical/biosafety reasons.
4. **Strain-outcome link is associative.** The A-east/A-west severity difference is epidemiologic; the molecular determinants of the virulence difference are not fully defined.

5. **Quantitative human phenotype frequencies and QoL metrics** specific to pneumonic tularemia are sparse; frequencies above are qualitative.
6. **No vaccine and limited chronic-disease data.** LVS is unlicensed and incompletely protective; the frequency and biology of chronic respiratory tularemia are based on rare case reports.

Proposed Follow-up Experiments / Actions

1. **Define the molecular basis of A-east vs A-west virulence** via comparative genomics/transcriptomics of matched clade isolates in the murine aerosol model, correlating specific pseudogenes/SNPs with "stealth" duration and lethality.
2. **Human host-genetics study** — targeted or exome analysis of severe vs mild pneumonic tularemia cohorts to test inflammasome/IFN/TLR susceptibility loci.
3. **Therapeutic inflammasome/host-directed adjuncts** — test whether relieving early inflammasome repression (or timed IL-1/IFN modulation) improves outcomes when combined with antibiotics in the mouse model.
4. **Correlate-of-protection validation for next-gen vaccines** — benchmark defined Schu S4 deletion mutants (e.g., $\Delta fopC$, $\Delta clpB$) against aerosol Type A challenge using perforin/granzyme and IFN- γ /TNF- α T-cell readouts as go/no-go criteria.
5. **Environmental risk mapping** — extend the Martha's Vineyard salt-soil persistence finding to a broader survey predicting where aerosol/landscaping-associated pneumonic transmission can occur.
6. **Rapid point-of-care diagnostics** — advance validated *tul4/fopA* PCR and antigen assays to shorten the currently serology-dependent, retrospective diagnostic window.

Report compiled from 10 confirmed findings and 58 reviewed papers across 5 investigation iterations. Evidence types are annotated (human clinical, model organism, in vitro, genomic). Ontology suggestions provided for HPO, GO, CL, UBERON, NCIT, and NCBI Taxonomy where applicable.